Tutorial: RCC – HSI, PLL

I. Overview

In this lab, we will learn how to configure the system clock (RCC) for HSI and PLL clock. The register of Reset and Clock Control can configure the system clock speed, oscillator source and can turn it on or off.

The system clock(SYSCLK) has three options for the input oscillation: Internal oscillator(HSI), External oscillator(HSE) and PLL. For PLL, it has the source either HSI and HSE. In this tutorial, we will use only HSI(16MHz) and HSI PLL (84MHz).

The objectives of this lab are learning how to

- Configure registers of RCC
- Enable HSI System Clock
- Enable PLL System Clock

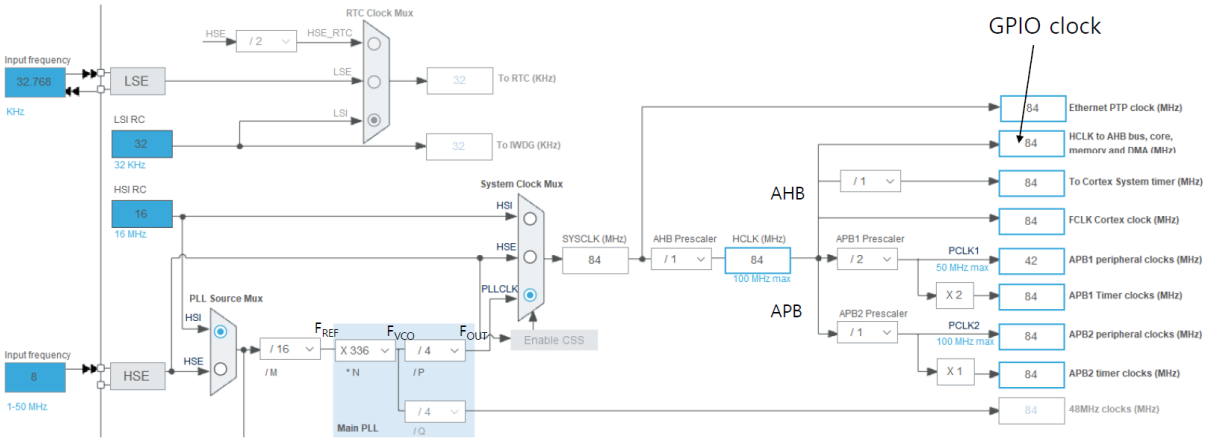
II. Tutorial

A. RCC Register

- List of RCC related registers for this tutorial

Type	Register Name	Description
RCC	**CR**	Enable the clock of GPIO Port
	CFGR **PLLCFGR**	Configuration System Clock Configuration PLL
	AHB1ENR	Peripheral(GPIO) clock

- RCC block diagram



B. Register Initialization Process for HSI(Internal Clock)

[RCC]

HSI as SYSCLK source

1. **Set HSI for HSI**

- Enable HSI: (RCC -> CR : HSION = 1)
- Wait until HSI is stable and ready: (RCC -> CR : HSIRDY ? 1)
- Choose the system clock switch : (RCC -> CFGR : SW = 00)
- Check if the selected source is correct: (RCC -> CFGR : SWS ? 00)

[GPIO]

2. **Enable GPIOx clock(AHB1ENR)**

- Enable (RCC_AHB1ENR) for PORTx

C. Register Initialization Process for PLL(HSI)

[RCC]

PLL as SYSCLK source

1. **Set HSI for PLL**

- Enable HSI (RCC -> CR : HSION = 1)
- Wait until HSI is stable (RCC -> CR : HSIRDY ? 1)

2. **Configure PLL parameters**

- Disable PLL for configuration (RCC -> CR : PLLON = 0)
- Select the clock source for PLL (RCC -> PLLCFGR : PLLSRC = 0) HSI
- Select (M/N/P): (RCC -> PLLCFGR : PLLM, PLLN, …

* (Option 1) M=16, N=336, P=4 // 16MHz*336/16/4 = 84MHz

* (Option 2) M=8, N=84, P=2 // 84MHz

3. **Enable PLL**
   - Enable PLL: (RCC -> CR : PLLON = 1)
   - Wait until PLL is stable: (RCC -> CR : PLLRDY ? 1)

4. **Set PLL for System CLOCK**
   - Choose PLL for system clock: (RCC -> CFGR : SW = 10)
   - Check if PLL selection is correct: (RCC -> CFGR : SWS ? 10)

Fill in the table

Register	Description	Register setting
RCC	Enable **HSI** (HSION = 1)	RCC->CR |= 1 << 0;
	Wait until **HSI** is stable and ready (HSIRDY == 1)	while ((RCC->CR >> 1) & 1 == 0) // not ready
	Choose the system clock switch (SW:00) for HSI	RCC->CFGR &=~ 0x3<<0 // clear bits RCC->CFGR |= 0 << 0;
	Check if the selected source is HSI (SWS:00)	while ((RCC->CFGR >> 2) & 0x3 != 0) // not correct
	Disable PLL for configuration (PLLON)	RCC -> CR & = ~1 << 24;
	Choose PLL for system clock (SW)	RCC -> CFGR &= ~3 << 0; RCC -> CFGR |= 2 << 0;
	Check if PLL selection is correct (SWS)	While ((RCC -> CFGR >> 2) & 3 != 2)
	Select the clock source for PLL (HSI) (PLLSRC)	RCC -> PLLCFGR &= ~1<< 22;
	Select (M/N/P): M=16, N=336, P=4	RCC -> PLLCFGR &= ~(63) << 0; RCC -> PLLCFGR |= 16 << 0; RCC -> PLLCFGR &= ~511 << 6; RCC -> PLLCFGR |= 336 << 6; RCC -> PLLCFGR &= ~3 << 16;

		RCC -> PLLCFGR |= 1 << 16;
	Enable PLL after configuration	RCC -> CR |= 1 << 24;
	Wait until PLL is stable	While ((RCC -> CR >> 25) & 1 != 0)
	Enable GPIOA Clock Enable GPIOx Clock (x = A~E)	RCC->AHB1ENR |= 1 << 0; RCC->AHB1ENR |= 1 << x;

6.3.1 RCC clock control register (RCC_CR)

Address offset: 0x00

Reset value: 0x0000 XX81 where X is undefined.

Access: no wait state, word, half-word and byte access

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved				PLLI2S RDY	PLLI2S ON	PLLRDY	PLLON	Reserved				CSS ON	HSE BYP	HSE RDY	HSE ON
				r	rw	r	rw					rw	rw	r	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HSICAL[7:0]								HSITRIM[4:0]					Res.	HSI RDY	HSION
r	r	r	r	r	r	r	r	rw	rw	rw	rw	rw		r	rw

6.3.3 RCC clock configuration register (RCC_CFGR)

Address offset: 0x08

Reset value: 0x0000 0000

Access: 0 ≤ wait state ≤ 2, word, half-word and byte access

1 or 2 wait states inserted only if the access occurs during a clock source switch.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MCO2		MCO2 PRE[2:0]			MCO1 PRE[2:0]			I2SSCR	MCO1		RTCPRE[4:0]				
rw		rw	rw	rw	rw	rw	rw	rw	rw		rw	rw	rw	rw	rw

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PPRE2[2:0]			PPRE1[2:0]			Reserved		HPRE[3:0]				SWS1	SWS0	SW1	SW0
rw	rw	rw	rw	rw	rw			rw	rw	rw	rw	r	r	rw	rw

6.3.9 RCC AHB1 peripheral clock enable register (RCC_AHB1ENR)

Address offset: 0x30

Reset value: 0x0000 0000

Access: no wait state, word, half-word and byte access.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved									DMA2EN	DMA1EN	Reserved				
									rw	rw					

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			CRCEN	Reserved				GPIOHEN	Reserved		GPIOEEN	GPIODEN	GPIOCEN	GPIOBEN	GPIOAEN
			rw					rw			rw	rw	rw	rw	rw

III. Exercise

A. Create EC_HAL functions (see ecRCC_student.c, .h)

Include File	Function	Description
ecRCC.h, c	void RCC_HSI_init(void);	HSI(16MHz)
	void RCC_PLL_init(void);	84MHz from HSI(16MHz) Select (M/N/P): M=16, N=336, P=4

	void RCC_GPIOA_enable(void);
	void RCC_GPIOB_enable(void);
	void RCC_GPIOC_enable(void);
	void RCC_GPIOD_enable(void);
	void RCC_GPIOE_enable(void);	